

Basic OSPF Routing Lab with 3 Routers and 2 PCs

Network Engineering Lab Portfolio

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1 Overview

This lab guide walks through configuring and verifying a single-area OSPF domain on Cisco IOS routers. It aims to establish dynamic routing across an Area 0 backbone, validate end-to-end network connectivity, and analyze OSPF neighbor state transitions using live console debugging.

2 Lab Prerequisites & Assumptions

- Basic understanding of IP addressing, wildcard masks, and CIDR subnetting.
- Physical interfaces and cabling are initialized and in an up/up operational state.
- Host devices (**PC1** and **PC2**) have default gateways pointed to their respective local router interfaces.

3 Network Topology & Subnet Scheme

The topology features three Cisco routers (**R1**, **R2**, **R3**) interconnected linearly, with two PCs (**PC1**, **PC2**) acting as host subnets attached to the perimeter routers[cite: 1].

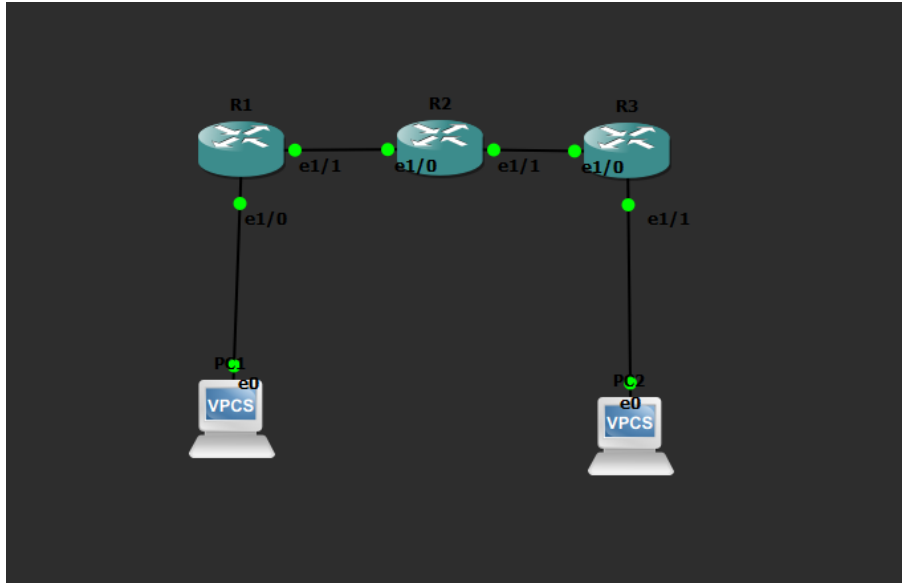


Figure 1: Lab Network Topology Diagram.

Device	Interface	IP Address	Subnet Mask	OSPF Area	Description
R1	Loopback0	1.1.1.1	255.255.255.255	Area 0	Router ID / Management Target
R1	Ethernet1/1	10.1.1.1	255.255.255.252	Area 0	Inter-router Link (to R2)[cite: 1]
R1	Ethernet1/0	192.168.10.1	255.255.255.0	Area 0	Default Gateway for PC1[cite: 1]
R2	Loopback0	2.2.2.2	255.255.255.255	Area 0	Router ID / Management Target
R2	Ethernet1/0	10.1.1.2	255.255.255.252	Area 0	Inter-router Link (to R1)
R2	Ethernet1/1	10.2.2.1	255.255.255.252	Area 0	Inter-router Link (to R3)
R3	Loopback0	3.3.3.3	255.255.255.255	Area 0	Router ID / Management Target
R3	Ethernet1/0	10.2.2.2	255.255.255.252	Area 0	Inter-router Link (to R2)
R3	Ethernet1/1	192.168.20.1	255.255.255.0	Area 0	Default Gateway for PC2
PC1	eth0	192.168.10.123	255.255.255.0	N/A	Host attached to R1
PC2	eth0	192.168.20.123	255.255.255.0	N/A	Host attached to R3

Table 1: Addressing and Role Schema.

3.1 Addressing Chart

4 Step-by-Step Configuration Guide

4.1 Step 1: Interface IP Address Configuration

Assign IP addresses to physical links, host gateways, and loopbacks across all routers.

Router R1:

```

R1(config)# interface Loopback0
R1(config-if)# ip address 1.1.1.1 255.255.255.255
R1(config-if)# interface Ethernet1/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# interface Ethernet1/1
R1(config-if)# ip address 10.1.1.1 255.255.255.252
R1(config-if)# no shutdown
  
```

Router R2:

```
R2(config)# interface Loopback0
R2(config-if)# ip address 2.2.2.2 255.255.255.255
R2(config-if)# interface Ethernet1/0
R2(config-if)# ip address 10.1.1.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# interface Ethernet1/1
R2(config-if)# ip address 10.2.2.1 255.255.255.252
R2(config-if)# no shutdown
```

Router R3:

```
R3(config)# interface Loopback0
R3(config-if)# ip address 3.3.3.3 255.255.255.255
R3(config-if)# interface Ethernet1/0
R3(config-if)# ip address 10.2.2.2 255.255.255.252
R3(config-if)# no shutdown
R3(config-if)# interface Ethernet1/1
R3(config-if)# ip address 192.168.20.1 255.255.255.0
R3(config-if)# no shutdown
```

4.2 Step 2: OSPF Area 0 Routing Setup

Enable the OSPF process, explicitly set manual Router IDs using software loopbacks, and declare wildcard masks for Area 0[cite: 1].

Router R1 Configuration:

```
R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 1.1.1.1 0.0.0.0 area 0
R1(config-router)# network 10.1.1.0 0.0.0.3 area 0
R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
```

Router R2 Configuration:

```
R2(config)# router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 2.2.2.2 0.0.0.0 area 0
R2(config-router)# network 10.1.1.0 0.0.0.3 area 0
R2(config-router)# network 10.2.2.0 0.0.0.3 area 0
```

Router R3 Configuration:

```
R3(config)# router ospf 1
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 3.3.3.3 0.0.0.0 area 0
R3(config-router)# network 10.2.2.0 0.0.0.3 area 0
R3(config-router)# network 192.168.20.0 0.0.0.255 area 0
```

5 Verification & Protocol Debugging

5.1 Step 3: Control Plane Verification

Run standard inspection commands across all routers to verify that remote loopbacks and host subnets correctly populate the routing table via OSPF (0)[cite: 1]:

Listing 1: R1 OSPF Routing Table Output

```
R1#show ip route ospf
      3.0.0.0/32 is subnetted, 1 subnets
0       3.3.3.3 [110/21] via 10.1.1.2, 00:09:29, Ethernet1/1
0      192.168.20.0/24 [110/30] via 10.1.1.2, 00:09:29, Ethernet1/1
      10.0.0.0/30 is subnetted, 2 subnets
0       10.2.2.0 [110/20] via 10.1.1.2, 00:09:29, Ethernet1/1
```

Listing 2: R2 OSPF Routing Table Output

```
R2#show ip route ospf
      3.0.0.0/32 is subnetted, 1 subnets
0       3.3.3.3 [110/11] via 10.2.2.2, 00:10:03, Ethernet1/1
0      192.168.10.0/24 [110/20] via 10.1.1.1, 00:10:03, Ethernet1/0
0      192.168.20.0/24 [110/20] via 10.2.2.2, 00:10:03, Ethernet1/1
```

Listing 3: R3 OSPF Routing Table Output

```
R3#show ip route ospf
0      192.168.10.0/24 [110/30] via 10.2.2.1, 00:09:02, Ethernet1/0
      10.0.0.0/30 is subnetted, 2 subnets
0       10.1.1.0 [110/20] via 10.2.2.1, 00:09:02, Ethernet1/0
```

5.2 Step 4: End-to-End Testing

Confirm end-to-end communication between the PCs across multiple routing hops:

```
PC1> ping 192.168.20.123
84 bytes from 192.168.20.123 icmp_seq=1 ttl=61 time=40.321 ms
84 bytes from 192.168.20.123 icmp_seq=2 ttl=61 time=36.887 ms
```

5.3 Step 5: Protocol Deep Dive (OSPF State Machine Analysis)

By initiating an interface restart (`shutdown` followed by `no shutdown`) while running `debug ip ospf adj`, we capture live OSPF state transitions on the console:

Listing 4: Live Console Debug Log

```
*Sep 15 21:58:10.978: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet1/1 from FULL
to DOWN
*Sep 15 21:58:27.202: OSPF: 2 Way Communication to 2.2.2.2 on Ethernet1/1, state 2WAY
*Sep 15 21:58:31.826: OSPF: NBR Negotiation Done. We are the SLAVE
*Sep 15 21:58:31.834: OSPF: Exchange Done with 2.2.2.2 on Ethernet1/1
*Sep 15 21:58:31.846: OSPF: Synchronized with 2.2.2.2 on Ethernet1/1, state FULL
*Sep 15 21:58:31.846: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet1/1 from
LOADING to FULL
```

- **2-WAY:** Hello packets are verified bidirectionally.
- **EXSTART:** Master/Slave roles are elected based on higher Router IDs.
- **EXCHANGE:** Summary Database Description (DBD) headers are exchanged.
- **LOADING:** Detailed Link State Requests (LSR) and Updates (LSU) sync missing entries.
- **FULL:** Both routers achieve identical Link State Databases (LSDB).